

Topology of the 24-Cell and F4 Symmetries in Quantum Field Theory

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We investigate the algebraic structure of the 24-cell polytope 3,4,3 and its automorphism group F4. By decomposing the 24-cell into three orthogonal 16-cell sub-lattices, we demonstrate the emergence of D4 triality. We establish how the gauge symmetries of the Standard Model arise naturally from Dynkin diagram deletions of the F4 root system, presenting a top-down geometric derivation of gauge groups in quantum field theory.

I. INTRODUCTION

The unification of gauge symmetries with gravity is a long-standing goal of mathematical physics. We approach this by studying the Dynkin diagram of F4, the exceptional Lie algebra associated with the symmetries of the 24-cell.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$24 = 8+8+8$ (decomposition into three orthogonal 16-cells)

(1)

$$F_4 \xrightarrow{\text{deletion}} SU(3)_C \times SU(2)_L \times U(1)_Y \quad (2)$$

III. DYNKIN DELETIONS

By removing peripheral Dynkin nodes of the F4 algebra, we isolate the subalgebra corresponding to the Standard Model gauge group, demonstrating that flavour triality is topological.

IV. CONCLUSION

The root system of the 24-cell uniquely determines the gauge structure of the early universe, precluding any additional dark gauge forces at the Planck scale.

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